

FLEX-LIVING

Business Requirements Document (BRD) | Product Requirements Document (PRD) | Technical Specification

Version 2.0 — Fused Strategic & Technical Build Plan

Classification: Confidential — Engineering, Product & Operations Reference

Target Markets: Accra (Ghana) -> Lagos (Nigeria) -> Nairobi (Kenya) -> Pan-African

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1. EXECUTIVE SUMMARY

1.1 The Opportunity

The African urban rental market is a \$180+ billion addressable market impaired by systemic trust failures:

- Advance rent extortion: 18-24 months upfront in West Africa (vs. 1 month globally)
- Predatory middlemen: 10-20% commission on total advance amounts
- Listing fraud: 70% of online listings in Nigeria are fake or fraudulent
- Infrastructure opacity: No verification of power, internet, water, or security before move-in

- Diaspora exclusion: \$95B+ in annual remittances with no trusted property management rail

1.2 The Solution

Flex-Living is Africa's first verified living infrastructure platform – not a rental marketplace, but a trust-and-financial layer that makes renting in Africa safe, flexible, and fair.

Core Value Propositions

Stakeholder	Promise	Mechanism
Tenants	"Rent without the burden"	Flex-Advance financing, verified badges, automated SLA protection
Landlords	"List free, earn more"	Zero commission, verified tenant matching, automated rent collection
Flex Guides (Scouts)	"Earn from your community"	Per-verification pay + match commissions + community status
Diaspora Investors	"Manage Africa from anywhere"	USDC rails, automated escrow, tax compliance, verified badges
Employers	"Housing as a benefit"	Payroll deduction integration, employee retention tool

1.3 The Fusion Thesis

This document fuses two strategic inputs:

1. Deep market strategy – Community-driven growth, offline-first design, zero-commission model, RentSusu financing, pan-African expansion
2. Technical PRD v1.0 – Microservices architecture, cryptographic scout audits, infrastructure badging, automated SLA escrow, IoT telemetry, multi-rail payments including USDC stablecoins

The fused product creates 7 competitive moats no single competitor can replicate.

1.4 Investment Ask

Phase	Timeline	Capital Required	Milestone
Seed	Months 0-6	\$350K	500 verified listings, 1,000 users, 20 Flex Guides, Accra only
Series A	Months 6-18	\$2.5M	50,000 users, 3 cities, Flex-Advance live, profitable unit economics
Series B	Months 18-36	\$8M	200,000 users, 8 countries, API ecosystem, market leadership

2. MARKET CONTEXT & PROBLEM ANALYSIS

2.1 Market Size

Metric	Value	Source
African urban population	600M+ (2026), 1B by 2040	UN-Habitat
Annual remittances to Africa	\$95B+	World Bank
Ghana rental market	\$2.5B annual	Ghana Statistical Service
Lagos rental market	\$8B annual	Lagos Bureau of Statistics
Nairobi rental market	\$3B annual	Kenya National Bureau of Statistics
Informal rental sector	80% of housing stock	African Housing Research

2.2 The Six Fractures

Fracture 1: The Advance Rent Prison

- West Africa: 18-24 months upfront (Ghana Rent Act 1963 caps at 6 months – zero enforcement)
- Nigeria: 1-2 years standard in Lagos, Abuja
- Human cost: Tenants sell assets, withdraw children from school, or take 40% interest loans

Fracture 2: The Middleman Tax

- Agent fees: 10% of total advance (not monthly rent)
- For a N2M/year apartment with 2-year advance: N400,000 agent fee alone
- 80% of agents in Ghana unlicensed despite REAC Act 2020

Fracture 3: The Trust Collapse

- 70% of Nigerian online rental listings are fraudulent
- AI deepfakes, cloned WhatsApp business profiles, phantom properties
- Tenants lose life savings to scams with zero recourse

Fracture 4: Infrastructure Roulette

- “Does the power work?” is asked before “how many bedrooms?”
- No verification of generator capacity, internet speed, water backup
- Post-move-in disputes are the norm, not the exception

Fracture 5: The Informal Shadow

- 80% of rentals are informal: no contracts, cash payments, no receipts
- 95% of landlords don’t record payments
- Tenants have zero credit history from rental payments

Fracture 6: Diaspora Exclusion

- Diaspora investors face 30% withholding tax, 10% management fees, fraud risk
- Net yields collapse to 2-4% after all costs

- No trusted platform for remote property management

2.3 Competitive Landscape

Competitor	Market	Strength	Weakness	Flex-Living Advantage
PropertyPro.ng	Nigeria	Volume leader (65% share)	Zero verification, agent-dependent	Cryptographic verification + zero commission
BuyRentKenya	Kenya	Traffic leader	No agent licensing, shallow verification	Full KYC + infrastructure badging
Ownkey	Ghana	Best-in-class verification	No financing, no diaspora rails	Flex-Advance + USDC + automated SLA
Airbnb	Global	Brand recognition	No MoMo, high fees, no advance financing	Local payments + financing + long-term focus
Jiji/OLX	Multi-market	Classifieds reach	Zero trust infrastructure	End-to-end verification + escrow

2.4 Target Personas

Primary Persona: “Ama the Young Professional” (Accra)

- 26, software developer, earns C4,000/month
- Needs to move closer to work in Cantonments
- Faces C48,000 (12-month) advance demand
- Uses MTN MoMo, WhatsApp, Instagram
- Pain: “I have the monthly income but not the advance. And I don’t trust agents.”

Secondary Persona: “Chidi the Diaspora Investor” (London/Lagos)

- 42, investment banker, owns 3 properties in Lekki
- Manages remotely through unreliable cousins/agents
- Receives rent inconsistently, suspects under-collection
- Pain: “I don’t know if my tenant is real, if the rent is full, or if my cousin is stealing.”

Tertiary Persona: “Kofi the Landlord” (Madina)

- 55, retired teacher, owns 2 rental units
- Uses basic Nokia + WhatsApp on son’s phone
- Wants reliable tenants, hates agent fees
- Pain: “Agents bring me tenants who default. I want someone who pays and stays.”

3. PRODUCT VISION & POSITIONING

3.1 Vision Statement

“Every African deserves a safe, verified place to live – without paying two years upfront to a stranger on WhatsApp.”

3.2 Mission

To build Africa’s trust infrastructure for housing: verified listings, flexible financing, automated protection, and community-powered growth.

3.3 Brand Positioning

Element	Specification
Name	Flex-Living
Tagline	“Rent Without the Burden”
Tone	Direct, warm, empowering, slightly irreverent
Colors	Coral (#E94560) + Deep Teal (#0F3460) + Gold (#E9A319)
Mascot	“Ama the Flex Guide” –friendly, community-rooted, tech-savvy
Brand Promise	Verified. Flexible. Fair.

3.4 The Five Pillars

Pillar 1: Flex-Trust

“Every listing is cryptographically verified by a human you can meet.”

Pillar 2: Flex-Advance

“Pay monthly, live immediately – no more 2-year ransom.”

Pillar 3: Flex-Stay

“If the power dies, we move you and refund you. Automatically.”

Pillar 4: Flex-Move

“From your old door to your new bed in one tap.”

Pillar 5: Flex-Shield

“The platform governments wish they’d built.”

4. BUSINESS REQUIREMENTS (BRD)

4.1 Revenue Model

Revenue Stream	Description	Margin	Phase
Flex-Advance Interest Spread	5-8% APR on rent advance financing	60% gross	Phase 2
Flex-Move Commission	15% on logistics bookings (movers, cleaners, vans)	15% net	Phase 1
Diaspora Management Fee	5% of monthly rent (vs. 10% traditional agents)	80% gross	Phase 3
Premium Listings	Featured placement, analytics dashboard	\$9.99/mo	Phase 2
Micro-Insurance	Rent protection, goods-in-transit, deposit insurance	20% net	Phase 2
B2B Corporate Housing	Per-employee fee for seamless relocation	\$50/employee/mo	Phase 3
API Licensing	Third-party fintechs querying Flex Score data	\$0.10/query	Phase 4

4.2 Unit Economics (Month 18 Projection)

Metric	Value
Customer Acquisition Cost (tenant)	\$3.50 (organic/community-driven)
Customer Acquisition Cost (landlord)	\$8.00 (scout-driven)
Lifetime Value (tenant, 24-month stay)	\$180 (Flex-Move + insurance + referrals)
Lifetime Value (landlord, 5-year relationship)	\$420 (premium listings + diaspora fees)
LTV:CAC Ratio	25:1 (tenant), 52:1 (landlord)
Gross Margin (platform level)	68%
Payback Period	4 months

4.3 Go-to-Market Strategy

Phase 1: Trust Seed (Months 0-6) – Accra, Ghana

Objective: Prove the trust model works at small scale.

Tactic	Execution	Budget	Target
Flex Guide Recruitment	20 guides from church/mosque/market networks	\$2,000/mo	500 verified listings
Zero-Commission Launch	First 100 landlords list free forever	\$0	100 direct hosts
University Ambassadors	10 campus reps at University of Ghana, KNUST, etc.	\$1,500/mo	1,000 student tenants
WhatsApp Bot	Property listing via WhatsApp message	\$500 dev	200 listings via bot
Church/Mosque Partnerships	20 congregations, sponsored announcements	\$1,000/mo	500 community members

Phase 2: Liquidity Burst (Months 6-12) – Ghana-Wide

Objective: Scale listings and launch financing.

Tactic	Execution	Target
Flex-Advance Pilot	100 rotating savings circles	1,000 financed moves
Employer Partnerships	3 corporate payroll deduction deals	500 salaried tenants
Market Women Network	50 market association chairs as referral nodes	2,000 organic signups
Scout Expansion	50 guides across Kumasi, Takoradi, Tamale	5,000 listings

Phase 3: Financial Lock-in (Months 12-18) – Lagos + Nairobi

Objective: Prove cross-border model and capture diaspora.

Tactic	Execution	Target
Lagos Launch	30 Flex Guides in Lekki, Ikeja, Yaba	15,000 listings
Nairobi Launch	M-Pesa Daraja integration, 20 guides	10,000 listings
Diaspora “Flex-Living Global”	USDC rails, London/NYC marketing	500 diaspora landlords
Tax Filing Integration	Auto GRA/FIRS/KRA reporting	1,000 compliant landlords

Phase 4: Pan-African Scale (Months 18-36) – 8 Countries

Objective: Market leadership and profitability.

Tactic	Execution	Target
API Ecosystem	Open Flex Score API to 20 fintech partners	50,000 API calls/day
Regulatory Partnerships	Licensed/endorsed in 8 markets	Government contracts
IoT at Scale	5,000 smart plugs deployed	Automated telemetry network
Profitability	Break-even on operations	\$500K monthly net income

4.4 Partnership Requirements

Partner Type	Examples	Value Exchange
Mobile Money Operators	MTN MoMo, M-Pesa, Telecel Cash, AirtelTigo	Revenue share on transaction volume; co-marketing
Payment Aggregators	Paystack, Flutterwave	Lower processing rates in exchange for volume commitment
Identity Verification	Smile Identity, Persona, Ghana Card API	KYC automation
Stablecoin Infrastructure	Circle (USDC), Solana Foundation	Diaspora remittance rails

Table 12 – continued

Partner Type	Examples	Value Exchange
Corporate HR Platforms	Workday, SeamlessHR, local payroll providers	Employee housing benefit integration
Insurance Partners	MicroEnsure, BIMA, local underwriters	White-labeled rent protection
Logistics Providers	Local van/truck owner associations	On-demand moving fleet
Government	REAC (Ghana), FIRS (Nigeria), KRA (Kenya)	Data sharing for policy; tax compliance automation

5. PRODUCT REQUIREMENTS (PRD)

5.1 Product Hierarchy

Flex-Living Platform

- Guest/Tenant App (React Native / Flutter)
- Search & Discovery
- Badge-Filtered Browse
- Flex-Advance Enrollment
- Booking & Escrow
- Stay Management (SLA Telemetry)
- 1-Tap Relocation
- Flex-Move Booking
- Host/Landlord App (React Native / Flutter + Web Dashboard)
- Listing Creation (App + WhatsApp Bot + USSD)
- Badge Management
- IoT Telemetry Dashboard
- Tenant Screening (Flex Score)
- Rent Collection & Payouts
- Tax Reporting
- Flex Guide App (React Native / Flutter)
- Audit Ticket Dispatch
- Offline Data Capture
- Cryptographic Signing
- Badge Assignment

- Earnings Dashboard
- Diaspora Dashboard (Web)
- Portfolio Management
- USDC Deposit/Withdrawal
- Tenant Screening
- Automated Tax Filing
- Performance Analytics
- Admin & Operations (Web)
- Scout QA & Dispatch
- Dispute Arbitration
- Financial Reconciliation
- Market Analytics
- Public API Layer
- Flex Score Query
- Badge Verification
- Listing Search

5.2 User Stories

Guest/Tenant User Stories

ID	Story	Priority	Acceptance Criteria
US-G-01	As a tenant, I want to search for rooms by infrastructure badges so that I know the power and internet work before I visit	P0	Badge filters functional; results load <2s; badge icons visible on cards
US-G-02	As a tenant, I want to enroll in Flex-Advance so that I can pay monthly instead of 2 years upfront	P0	Circle creation <5 min; payment schedule clear; employer verification option

Table 13 – continued

ID	Story	Priority	Acceptance Criteria
US-G-03	As a tenant, I want to book a verified listing with escrow protection so that my money is safe until I move in	P0	Escrow hold confirmed via SMS; digital lease generated; 24-hr post-check-in release
US-G-04	As a tenant, I want to report an infrastructure outage with one tap so that I get compensated automatically	P1	Report categories: Power, Wi-Fi, Water, Lock; auto-diagnostic runs; cure timer starts
US-G-05	As a tenant, I want to request relocation if my stay is compromised so that I don' t sleep without power	P1	1-tap relocation triggers; equivalent properties suggested; move dispatched
US-G-06	As a tenant, I want to book a moving van through the app so that I don' t negotiate with random drivers	P1	Fixed pricing by distance; GPS tracking; driver verification visible
US-G-07	As a tenant, I want to build my Flex Score so that I qualify for better properties and faster advances	P2	Score visible on profile; improvement tips; history of payments and reviews
US-G-08	As a tenant without smartphone data, I want to use USSD to check listings and make payments	P2	USSD code 1231# functional; basic search and payment flows work

Host/Landlord User Stories

ID	Story	Priority	Acceptance Criteria
US-H-01	As a landlord, I want to list my property via WhatsApp so that I don' t need to download an app	P0	Send photos + location to WhatsApp number; AI generates draft; scout confirms
US-H-02	As a landlord, I want to get verified badges for my property so that I attract more tenants	P0	Scout audit scheduled within 48hrs; badges displayed on listing; re-audit annually
US-H-03	As a landlord, I want to receive rent automatically via mobile money so that I don' t chase tenants	P0	Auto-debit set up; payment confirmation SMS; monthly statement generated
US-H-04	As a landlord, I want to see my property' s infrastructure telemetry so that I know when maintenance is needed	P1	IoT dashboard: power status, internet speed, water levels; alert thresholds
US-H-05	As a landlord, I want to auto-file my rental income taxes so that I stay compliant	P2	Tax report generated quarterly; one-tap filing to GRA/FIRS/KRA; payment confirmation

Flex Guide User Stories

ID	Story	Priority	Acceptance Criteria
US-S-01	As a Flex Guide, I want to receive audit tickets near me so that I can earn from local properties	P0	GPS dispatch within 5km radius; ticket details: address, badges requested, pay
US-S-02	As a Flex Guide, I want to capture audit data offline so that poor connectivity doesn't stop me	P0	App works offline; data syncs when connected; photos cached locally
US-S-03	As a Flex Guide, I want to cryptographically sign my audit report so that it can't be tampered with	P0	Ed25519 key generated on device; signature appended to payload; hash verified server-side
US-S-04	As a Flex Guide, I want to track my earnings and withdraw via mobile money so that I get paid quickly	P1	Earnings dashboard real-time; withdrawal request <24hr processing; minimum C50

Diaspora Investor User Stories

ID	Story	Priority	Acceptance Criteria
US-D-01	As a diaspora investor, I want to deposit USDC to fund my property portfolio so that I avoid FX spreads	P1	USDC deposit address generated; confirmation <10 min; balance visible in USD and local currency

Table 16 – continued

ID	Story	Priority	Acceptance Criteria
US-D-02	As a diaspora investor, I want to see verified badge reports for my properties so that I know their condition	P1	Scout audit reports with photos, videos, cryptographic signatures; annual re-audit alerts
US-D-03	As a diaspora investor, I want to receive rent in USDC automatically so that I avoid banking delays	P1	Monthly auto-conversion from local currency to USDC; withdrawal to wallet <24hrs

5.3 Core User Flows

Flow 1: Tenant Discovery -> Booking -> Move-In

[Guest opens app / WhatsApp / USSD] | [Search: Location + Budget + Badge Filters] | [Browse Verified Listings with Badge Icons] | [View Listing: Photos, Scout Report, Badge Details, Flex Guide Profile] | [Request Viewing / Instant Book (if BADGE_SEC_SELF_CHECKIN)] | [Flex-Advance Option: “Can’t pay full advance? Enroll in Flex-Advance”] | – [Employer Verified] -> Instant approval, payroll deduction | – [Informal Worker] -> Join rotating circle, wait for payout | [Escrow Payment: First month + deposit held in Flex-Living escrow] | [Digital Lease Generated + Signed] | [Flex-Move Booking: Select date, get fixed quote, track driver] | [Check-In: Photo inventory, digital key handover, Wi-Fi credentials] | [24-Hour Confirmation Window -> Escrow released to host] | [Stay Management: SLA telemetry, outage reporting, 1-tap relocation]

Flow 2: Landlord Listing Creation

[Landlord sends “Hi” to Flex-Living WhatsApp] | [Bot: “Send photos of your property and your location”] | [Landlord sends 5 photos + WhatsApp location pin] | [AI generates draft listing: auto-tags rooms, estimates price from comparables] | [Bot: “Confirm these details? Reply YES or EDIT”] | [Landlord confirms] | [System dispatches nearest Flex Guide for verification audit] | [Flex Guide visits, captures data offline, uploads, cryptographically signs] | [Badges awarded, listing goes live with “Flex-Verified” status] | [Landlord receives dashboard link via SMS]

Flow 3: Automated SLA Dispute

[Tenant taps “Report Outage” -> Selects “Power”] | [App runs diagnostic: IoT smart plug ping + ambient light video + router status] | [If breach confirmed: Start cure timer (2 hours for Tier 1)] | [Auto-SMS/WhatsApp/IVR alert to host: “Power outage detected. Fix within 2 hours.”] | [Host uploads resolution proof or fixes issue] | [Timer expires without resolution] | [Escrow auto-releases: 100% refund of unused nights + free relocation credit] | [1-Tap Relocation: Show equivalent badged properties, auto-rebook, dispatch Flex-Move] | [Host badge stripped + \$50 fine + strike logged]

6. SYSTEM ARCHITECTURE & SERVICE DOMAINS

6.1 Architectural Principles

1. Microservices with clear boundaries – Each domain owns its data; communication via async events
2. Event-driven for resilience – Apache Kafka / AWS SQS for cross-service communication
3. Offline-first for Africa – Mobile apps cache locally; sync when connected
4. Multi-tenant by design – Single codebase, geo-located deployments
5. API-first – All functionality exposed via REST/gRPC for web, mobile, USSD, WhatsApp
6. Security by default – Encryption at rest and in transit; zero-trust network

6.2 Service Domain Map

CLIENT LAYER

- Tenant Mobile App
 - Landlord Mobile App + Web Dashboard
 - Flex Guide Mobile App
 - Diaspora Web Dashboard
 - Admin Web Dashboard
 - WhatsApp Bot
 - USSD Gateway | v API GATEWAY (Kong / AWS API Gateway / Cloudflare)
 - Rate limiting
 - Auth (JWT/OAuth2)
 - Request routing | v SERVICE DOMAINS (6 Microservices)
1. Identity & Trust Core (Auth/KYC)
 2. Listing & Badge Engine (Catalog/Search)
 3. Field Scout Service (Audits)
 4. Escrow & Multi-Rail Billing (Payments)
 5. SLA & Dispute Engine (Protection)
 6. Move-In & Logistics Concierge (Moving) | v MESSAGE BUS (Apache Kafka / AWS SNS-SQS)
- audit.completed -> Badge Engine
 - booking.confirmed -> Escrow Engine
 - outage.reported -> SLA Engine -> Escrow Engine
 - payment.received -> Escrow Engine -> Host Payout
 - scout.ticket.created -> Scout Dispatch | v DATA LAYER
 - PostgreSQL (Primary DB + PostGIS)

- Redis (Cache + Session)
- S3 / MinIO (Media/Logs)
- Elasticsearch (Search)

6.3 Service Domain Specifications

Domain 1: Identity & Trust Core

Attribute	Specification
Responsibility	User registration, KYC/AML, identity verification, role management
Key Entities	Users, KYC_Profiles, Identity_Documents, Role_Assignments
External Integrations	Ghana Card API, Nigeria NIN API, Kenya ID API, Smile Identity, Persona SDK
Tech Stack	Node.js / Go, PostgreSQL, Redis (session cache)
API Style	REST + gRPC for internal

KYC Tiers:

Tier	Verification Level	Access
Tier 0	Phone number only	Browse only
Tier 1	Government ID + selfie match	Book short stays (<30 days)
Tier 2	Tier 1 + utility bill / employer letter	Book long stays + Flex-Advance
Tier 3	Tier 2 + credit bureau check	High-value Flex-Advance + premium properties

Domain 2: Listing & Badge Engine

Attribute	Specification
Responsibility	Property catalog, dynamic pricing, infrastructure badges, spatial search
Key Entities	Listings, Units, Badges, Pricing_Rules, GeoJSON_Indexes
External Integrations	What3Words API, Google Plus Codes, Google Maps API, Elasticsearch
Tech Stack	Python (Django/FastAPI), PostgreSQL + PostGIS, Elasticsearch

Badge System:

Category	Badge ID	SLA Standard	Verification Method
Power	BADGE_POWER_247_	UNBROKEN hybrid, <10s switchover, 100% load	Breaker trip test, 15s video, nameplate kVA validation
Power	BADGE_POWER_WORK_READY	15-30 min start sustaining Wi-Fi, lights, PC charging indefinitely	Load testing, LiFePO4 bank verification
Power	BADGE_POWER_ON_DEMAND_GEN	15-30 min start	Fuel reserve log, host protocol verification
Internet	BADGE_NET_ULTRA_FIBRE	100% up / 50 Mbps up, <25ms latency, uncapped	Dual speed test (router + dead zone), ISP billing OCR
Internet	BADGE_NET_PRO_WIRELESS	99.9% up / 30 Mbps up, <60ms latency, router on UPS	Speed test, router power source inspection
Security	BADGE_SEC_FORTIFIED_ESTATE	24/7 manned gate, electric fence, access control, zero indoor cameras	Gate interview, fence voltage check, camera privacy sweep
Security	BADGE_SEC_SELF_CHECKIN	High N-keypad/smart lockbox with dynamic OTP	Test code generation, remote unlock test

Domain 3: Field Scout Service

Attribute	Specification
Responsibility	Audit ticket dispatch, GPS geofencing, hardware telemetry, cryptographic signing
Key Entities	Audit_Tickets, Scout_Profiles, Audit_Payloads, Badge_Assignments
External Integrations	GPS/GNSS, Camera EXIF libraries, Ed25519 crypto libraries
Tech Stack	Go (high concurrency), SQLite (offline cache on device), AWS S3 (media)

Scout Workflow:

1. System creates audit ticket -> Pushed to nearest available scouts
2. Scout accepts ticket -> GPS navigation to What3Words address
3. Scout arrives -> Geofence verification (must be within 50m of listing)
4. Scout captures: Photos (EXIF hashed), Videos (trip test), Speed tests
5. Scout completes checklist -> App generates JSON audit payload
6. Scout signs payload with Ed25519 private key (stored in secure enclave)

7. Payload + signature uploaded to server (syncs when online)
8. Server verifies signature against scout's public key
9. Badge Engine processes payload -> Awards/denies badges
10. Listing updated with new badge status

Domain 4: Escrow & Multi-Rail Billing

Attribute	Specification
Responsibility	Payment collection, escrow state holding, split payouts, currency conversion, payroll deductions
Key Entities	Transactions, Escrow_States, Payouts, Currency_Conversions, Payroll_Deductions
External Integrations	MTN MoMo API, Telecel Cash, M-Pesa Daraja, Paystack, Flutterwave, Circle (USDC), Solana/Polygon RPC
Tech Stack	Java / Kotlin (financial transactions), PostgreSQL, Redis (idempotency keys)

Escrow State Machine: [Payment Received] | [HELD] -> [RELEASED] (24hrs post check-in, no dispute) | |- [DISPUTE_FROZEN] -> [RESOLVED_RELEASE] or [RESOLVED_REFUND] | |- [SLA_BREACH] -> [AUTO_REFUND] -> [RELOCATION_TRIGGERED] | |- [CANCELLED_BY_GUEST] -> [PARTIAL_REFUND] (per cancellation policy)

Payment Rails:

Rail	Markets	Use Case	Settlement Time
MTN MoMo	Ghana, Nigeria, 15+ countries	Primary tenant payment	Instant
M-Pesa Daraja	Kenya, Tanzania	Primary tenant payment	Instant
Telecel Cash	Ghana	Alternative mobile money	Instant
AirtelTigo Money	Ghana	Alternative mobile money	Instant
Paystack	Nigeria	Card/bank transfer	1-2 days
Flutterwave	Multi-African	Card/bank transfer	1-2 days
USDC (Solana/Polygon)	Global	Diaspora deposits/withdrawals	<10 min
Employer Payroll	Corporate	Salary deduction	Monthly

Domain 5: SLA & Dispute Engine

Attribute	Specification
Responsibility	Live diagnostics, automated outage countdowns, instant partial refunds, relocation routing

Table 24 – continued

Attribute	Specification
Key Entities	Outage_Claims, SLA_Timers, Refund_Records, Relocation_Bookings
External Integrations	Twilio (SMS/IVR), Africa' s Talking (SMS), IoT Smart Plug APIs, Speedtest SDK
Tech Stack	Node.js (event-driven), Redis (timer management), Kafka (event streaming)

Outage Tiers & Compensation:

Tier	Trigger	Cure Window	Guest Remedy	Host Penalty
Tier 1: Critical	Total power blackout >1hr; no water >3hrs; total internet >4hrs	2 hours	Relocation + 100% refund of unused nights + free ride credit	Badge stripped; \$50 fine; 1 strike
Tier 2: Degraded	AC offline during cut (24/7 badge); Wi-Fi <50% of badge tier >3hrs	4 hours	30% daily rate refund per affected 24hr cycle	Badge downgraded; warning strike
Tier 3: Minor	Generator start delay 15-45 min; smart lock battery depleted; hot water failure	6 hours	15% daily rate refund for that day	Maintenance receipt required within 48hrs

Domain 6: Move-In & Logistics Concierge

Attribute	Specification
Responsibility	On-demand moving dispatch, condition inventory check-in/out, utility token auto-handover
Key Entities	Move_Jobs, Vehicle_Fleets, Condition_Reports, Utility_Tokens
External Integrations	Google Maps Routing, ECG/Ikeja Electric/KPLC token APIs, local van owner networks
Tech Stack	Python (optimization), PostgreSQL, Redis (job queue)

7. DATA MODELS & DATABASE SCHEMA

7.1 Entity Relationship Overview

[Users] 1—* [KYC_Profiles] [Users] 1—* [Listings] (as Host) [Users] 1—* [Bookings] (as Guest) [Users] 1—* [Scout_Profiles] (as Flex Guide) [Users] 1—* [Flex_Advance_Circles]

[Listings] 1—* [Units] [Listings] 1—* [Audit_Reports] [Listings] 1—* [Badges] [Listings] 1—* [IoT_Devices]

[Units] 1—* [Bookings] [Units] 1—* [Pricing_Rules]

[Bookings] 1—1 [Escrow_States] [Bookings] 1—* [Payments] [Bookings] 1—* [Outage_Claims] [Bookings] 1—1 [Move_Jobs]

[Audit_Reports] 1—* [Badge_Assignments] [Audit_Reports] 1—1 [Cryptographic_Signatures]

[Flex_Advance_Circles] — [Users] (members) [Flex_Advance_Circles] 1—* [Circle_Payments]

7.2 Core Table Specifications

Table: users

Column	Type	Constraints	Description
id	UUID	PK	Unique identifier
phone_number	VARCHAR(20)	UNIQUE, NOT NULL	Primary login credential
email	VARCHAR(255)	UNIQUE, NULLABLE	Optional email
full_name	VARCHAR(255)	NOT NULL	Display name
role	ENUM	NOT NULL	guest, host, scout, admin, diaspora
kyc_tier	INT	DEFAULT 0	0=none, 1=basic, 2=verified, 3=premium
flex_score	INT	DEFAULT 300	300-850 rental credit score
preferred_language	VARCHAR(10)	DEFAULT 'en'	en, tw, ga, yo, sw, etc.
whatsapp_opt_in	BOOLEAN	DEFAULT TRUE	Marketing consent
created_at	TIMESTAMP	DEFAULT NOW()	Registration date
updated_at	TIMESTAMP	AUTO UPDATE	Last modified

Table: listings

Column	Type	Constraints	Description
id	UUID	PK	Unique identifier
host_id	UUID	FK -> users	Property owner
title	VARCHAR(255)	NOT NULL	Listing headline
description	TEXT	NULLABLE	Detailed description
property_type	ENUM	NOT NULL	apartment, house, room, studio, shared
address_w3w	VARCHAR(100)	NOT NULL	What3Words address
geo_location	GEOGRAPHY(POINT)	NOT NULL	PostGIS point
city	VARCHAR(100)	NOT NULL	Accra, Lagos, Nairobi, etc.
country	VARCHAR(100)	NOT NULL	Ghana, Nigeria, Kenya, etc.
base_price_daily	DECIMAL(10,2)	NULLABLE	Nightly rate (short stays)
base_price_monthly	DECIMAL(10,2)	NOT NULL	Monthly rent
advance_months_required	INT	DEFAULT 12	Upfront months demanded by host

Table 28 – continued

Column	Type	Constraints	Description
status	ENUM	DEFAULT ‘draft’	draft, pending_audit, live, paused, suspended
scout_id	UUID	FK -> users	Assigned Flex Guide
audit_date	TIMESTAMP	NULLABLE	Last audit timestamp
created_at	TIMESTAMP	DEFAULT NOW()	

Table: badges

Column	Type	Constraints	Description
id	UUID	PK	
listing_id	UUID	FK -> listings	
badge_type	ENUM	NOT NULL	power, internet, security, water, access
badge_id	VARCHAR(50)	NOT NULL	BADGE_POWER_247_UNBROKEN, etc.
awarded_at	TIMESTAMP	NOT NULL	
expires_at	TIMESTAMP	NOT NULL	Annual re-audit required
audit_report_id	UUID	FK -> audit_reports	Source audit
is_active	BOOLEAN	DEFAULT TRUE	

Table: audit_reports

Column	Type	Constraints	Description
id	UUID	PK	
listing_id	UUID	FK -> listings	
scout_id	UUID	FK -> users	Flex Guide who performed audit
status	ENUM	DEFAULT ‘pending’	pending, completed, rejected, appealed
payload_json	JSONB	NOT NULL	Full audit data per PRD schema
payload_hash	VARCHAR(64)	NOT NULL	SHA-256 of payload
scout_signature	VARCHAR(128)	NOT NULL	Ed25519 signature
scout_public_key	VARCHAR(128)	NOT NULL	Public key for verification
geofence_verified	BOOLEAN	NOT NULL	Within 50m of listing
gps_lat	DECIMAL(10,8)	NOT NULL	
gps_lng	DECIMAL(11,8)	NOT NULL	
created_at	TIMESTAMP	DEFAULT NOW()	

Table: bookings

Column	Type	Constraints	Description
id	UUID	PK	

Table 31 – continued

Column	Type	Constraints	Description
guest_id	UUID	FK -> users	Specific room/unit
listing_id	UUID	FK -> listings	
unit_id	UUID	FK -> units	
check_in	DATE	NOT NULL	
check_out	DATE	NOT NULL	
stay_type	ENUM	NOT NULL	
total_amount	DECIMAL(12,2)	NOT NULL	Total contract value
advance_amount	DECIMAL(12,2)	NOT NULL	Upfront payment
monthly_rent	DECIMAL(10,2)	NOT NULL	Recurring amount
flex_advance_enrolled	BOOLEAN	DEFAULT FALSE	Using Flex-Advance financing
escrow_state	ENUM	DEFAULT 'pending_payment'	See state machine
status	ENUM	DEFAULT 'pending'	pending, confirmed, active, completed, cancelled
digital_lease_url	VARCHAR(255)	NULLABLE	Signed contract
created_at	TIMESTAMP	DEFAULT NOW()	

Table: escrow_states

Column	Type	Constraints	Description
id	UUID	PK	held, released, dispute_frozen, auto_refund, cancelled
booking_id	UUID	FK -> bookings	
state	ENUM	NOT NULL	
amount_held	DECIMAL(12,2)	NOT NULL	
currency	VARCHAR(3)	NOT NULL	
held_at	TIMESTAMP	NOT NULL	
release_conditions	JSONB	NOT NULL	Conditions for release
released_at	TIMESTAMP	NULLABLE	FK -> users (host)
released_to	UUID	NULLABLE	
dispute_reason	VARCHAR(255)	NULLABLE	
resolved_at	TIMESTAMP	NULLABLE	

Table: flex_advance_circles

Column	Type	Constraints	Description
id	UUID	PK	e.g., “Accra East Circle 001”
name	VARCHAR(100)	NOT NULL	
target_amount	DECIMAL(12,2)	NOT NULL	
member_count	INT	NOT NULL	
contribution_amount	DECIMAL(10,2)	NOT NULL	
rotation_months	INT	NOT NULL	

Table 33 – continued

Column	Type	Constraints	Description
status	ENUM	DEFAULT 'forming'	forming, active, completed, defaulted
employer_id	UUID	NULLABLE	FK -> employers (if corporate)
default_insurance_paid	BOOLEAN	DEFAULT FALSE	5% of pool
created_at	TIMESTAMP	DEFAULT NOW()	

Table: payments

Column	Type	Constraints	Description
id	UUID	PK	
booking_id	UUID	FK -> bookings	NULLABLE (can be circle payment)
circle_id	UUID	FK -> flex_advance_circles	NULLABLE
user_id	UUID	FK -> users	Payer
amount	DECIMAL(12,2)	NOT NULL	
currency	VARCHAR(3)	NOT NULL	
payment_rail	ENUM	NOT NULL	mtn_momo, mpesa, telecel, paystack, flutterwave, usdc, payroll
external_transaction_id	VARCHAR(255)	NULLABLE	Provider reference
status	ENUM	NOT NULL	pending, success, failed, refunded
payment_type	ENUM	NOT NULL	advance, monthly, deposit, service_fee, insurance
created_at	TIMESTAMP	DEFAULT NOW()	

Table: outage_claims

Column	Type	Constraints	Description
id	UUID	PK	
booking_id	UUID	FK -> bookings	
guest_id	UUID	FK -> users	
category	ENUM	NOT NULL	power, internet, water, security, access
severity_tier	INT	NOT NULL	1, 2, or 3
description	TEXT	NULLABLE	Guest description
diagnostic_data	JSONB	NOT NULL	Speed test, IoT ping, video, etc.
cure_timer_started_at	TIMESTAMP	NOT NULL	
cure_timer_hours	INT	NOT NULL	2, 4, or 6
host_notified_at	TIMESTAMP	NOT NULL	
resolved_at	TIMESTAMP	NULLABLE	
resolution_proof	VARCHAR(255)	NULLABLE	Photo/video URL
auto_refund_triggered	BOOLEAN	DEFAULT FALSE	

Table 35 – continued

Column	Type	Constraints	Description
refund_amount	DECIMAL(10,2)	NULLABLE	
relocation_triggered	BOOLEAN	DEFAULT FALSE	

Table: move_jobs

Column	Type	Constraints	Description
id	UUID	PK	
booking_id	UUID	FK -> bookings	
guest_id	UUID	FK -> users	
pickup_address	VARCHAR(255)	NOT NULL	
pickup_w3w	VARCHAR(100)	NOT NULL	
delivery_address	VARCHAR(255)	NOT NULL	
delivery_w3w	VARCHAR(100)	NOT NULL	
vehicle_type	ENUM	NOT NULL	bike, van, truck
estimated_cost	DECIMAL(10,2)	NOT NULL	Fixed price
driver_id	UUID	NULLABLE	FK -> users (vetted driver)
status	ENUM	DEFAULT 'pending'	pending, dispatched, in_transit, completed, cancelled
gps_tracking_url	VARCHAR(255)	NULLABLE	Real-time tracking
condition_photos_before	JSONB	NULLABLE	Check-in inventory
condition_photos_after	JSONB	NULLABLE	Check-out inventory
created_at	TIMESTAMP	DEFAULT NOW()	

8. API SPECIFICATION

8.1 API Design Principles

- RESTful for external clients (mobile apps, web, WhatsApp bot)
- gRPC for internal service communication
- Versioned (/v1/, /v2/)
- Idempotent for payment operations (idempotency-key header required)
- Rate-limited per user tier (Tier 0: 100 req/min, Tier 1+: 1000 req/min)
- Localized (Accept-Language header: en, tw, ga, yo, sw)

8.2 Core Endpoints

Identity & Trust Core

POST /v1/auth/register # Phone OTP registration POST /v1/auth/verify-otp # OTP verification
 POST /v1/auth/refresh-token # JWT refresh POST /v1/kyc/submit # Submit ID + selfie GET /v1/kyc/status # Check verification status GET /v1/users/me # Get current user profile PATCH /v1/users/me # Update profile GET /v1/users/{id}/flex-score # Get public Flex Score

Listing & Badge Engine

GET /v1/listings # Search with filters GET /v1/listings/{id} # Get listing detail POST /v1/listings # Create listing (host) PATCH /v1/listings/{id} # Update listing DELETE /v1/listings/{id} # Deactivate listing GET /v1/listings/{id}/badges # Get active badges GET /v1/listings/{id}/audit-report # Get latest scout report POST /v1/listings/{id}/request-audit # Request new audit GET /v1/badges/definitions # Get all badge types and requirements

Search Query Parameters: GET /v1/listings? city=accra& min_price=500& max_price=2000& property_type=apartment& badges=BADGE_POWER_247_UNBROKEN,BADGE_NET_ULTRA_FIBRE_100& lat=5.6037&lng=-0.1870&radius_km=5& check_in=2026-09-01&check_out=2026-09-30& flex_advance_eligible=true& page=1&limit=20

Booking & Escrow

POST /v1/bookings # Create booking GET /v1/bookings/{id} # Get booking detail POST /v1/bookings/{id}/pay # Make payment to escrow POST /v1/bookings/{id}/cancel # Cancel booking POST /v1/bookings/{id}/confirm-checkin # Release escrow (guest) GET /v1/bookings/{id}/escrow # Get escrow status POST /v1/bookings/{id}/dispute # Raise dispute

Flex-Advance

POST /v1/flex-advance/circles # Create new circle GET /v1/flex-advance/circles # List available circles POST /v1/flex-advance/circles/{id}/join # Join circle GET /v1/flex-advance/circles/{id} # Get circle status POST /v1/flex-advance/circles/{id}/pay # Make monthly contribution GET /v1/flex-advance/eligibility # Check Flex-Advance eligibility POST /v1/flex-advance/employer-verify # Submit employer verification

SLA & Outages

POST /v1/outages/report # Report outage GET /v1/outages/{id}/status # Check claim status POST /v1/outages/{id}/resolve # Host marks resolved POST /v1/outages/{id}/relocation # Request relocation GET /v1/bookings/{id}/telemetry # Get live infrastructure telemetry

Move-In Logistics

POST /v1/moves/quote # Get moving quote POST /v1/moves/book # Book moving service GET /v1/moves/{id} # Track move POST /v1/moves/{id}/complete # Mark move complete

WhatsApp Bot Webhook

POST /v1/webhooks/whatsapp # Incoming WhatsApp messages POST /v1/webhooks/whatsapp/status # Delivery status updates

USSD Gateway

POST /v1/ussd/session # USSD session initiation POST /v1/ussd/response # USSD user input

8.3 WebSocket Events (Real-Time)

Event	Payload	Triggered By
booking.confirmed	Booking object + escrow status	Payment received
escrow.released	Amount, recipient, transaction ID	24hr post check-in or auto-resolution
outage.reported	Outage claim + cure timer	Guest report
outage.resolved	Resolution proof + refund amount	Host fix or timer expiry
scout.ticket.assigned	Ticket details, location, pay	System dispatch
move.dispatched	Driver details, ETA, tracking URL	Driver assignment

9. INFRASTRUCTURE BADGING & VERIFICATION MATRIX

9.1 Badge Lifecycle

[Listing Created] | [Audit Requested] -> Flex Guide dispatched | [Field Audit Completed] -> Cryptographic payload generated | [Payload Verified] -> Signature valid, geofence confirmed | [Badge Awarded] -> Added to listing record with expiry date | [Annual Re-Audit] -> 30 days before expiry, auto-request new audit | [Badge Expired] -> If no re-audit, badge removed, listing downgraded

9.2 Detailed Badge Specifications

Power Badges

Badge	Requirements	Test Protocol	Evidence Required
BADGE_POWER_247_UNBROKEN	ATS247 solution; <10s switchover; sustains 100% load including ACs	Main breaker trip test; 15s continuous video; nameplate kVA photo	Video URL, switchover time (ms), heavy loads sustained (boolean)
BADGE_POWER_WORK_READY	Eveready/Rechargeable; Wi-Fi router, PC charging, lights indefinitely (no AC)	Load test with ACs off; battery chemistry verification	Battery type, capacity (kWh), test duration
BADGE_POWER_ON_DEMAND_5_GEN	Manual generator; 5-min start; scheduled evening windows	Host protocol walkthrough; fuel reserve check; start test video	Fuel level %, generator capacity (kVA), start time (min)

Internet Badges

Badge	Requirements	Test Protocol	Evidence Required
BADGE_NET_ULTRAFAST	100Mbps down/50Mbps up; <25ms latency; uncapped fiber	Dual speed test: router line-of-sight + furthest workspace; ISP billing OCR	Two speed test results, ISP name, plan type, billing screenshot
BADGE_NET_PROWORK	30Mbps down/15Mbps up; <60ms latency; router on dedicated mini-UPS	Speed test at primary workspace; router power source inspection	Speed test result, router UPS present (boolean)

Security Badges

Badge	Requirements	Test Protocol	Evidence Required
BADGE_SEC_FORTIFIED_ESTATE	Perimeter gas, electric fence; access control; zero indoor cameras	Gate security interview; fence voltage monitor inspection; camera privacy sweep	Interview notes, voltage reading, camera sweep video
BADGE_SEC_SELF_CHECKIN	Digital keypack; smart lockbox; dynamic OTP generation	Test code generation; remote unlock test by scout	Lock type, test code generation success (boolean)

9.3 IoT Telemetry Integration

Device Type	Data Collected	Update Frequency	Badge Relevance
Smart Plug (Power)	On/off status, voltage, frequency	Every 60 seconds	BADGE_POWER_247_UNBRO
Speed Test Edge	Download/upload/latency/jitter	Every 5 minutes	BADGE_NET_*
Water Level Sensor	Tank capacity, current level	Every 15 minutes	Water badge (future)
Smart Lock	Battery level, unlock events	Event-driven	BADGE_SEC_SELF_CHECKIN

10. FIELD SCOUT ENGINE & CRYPTOGRAPHIC AUDIT

10.1 Scout Mobile App Specification

Platform: React Native (iOS/Android) with offline-first architecture

Core Features

Feature	Specification
Offline Cache	SQLite database stores tickets, photos, videos locally; syncs on connection
GPS Geofencing	Must be within 50m of listing coordinates to begin audit
Camera Capture	Native camera with EXIF preservation; automatic GPS tagging
Speed Test SDK	Custom in-app speed test (multi-threaded, tests to nearest cloud edge)
Cryptographic Signing	Ed25519 keypair generated in secure enclave; private key never leaves device
Breaker Trip Test	Video recording during main breaker flip; timer for switchover
WhatsApp Integration	Scout can message host directly for access coordination

Audit Payload Schema (JSON)

The full JSON schema is specified in the uploaded PRD document. Key required fields:

- audit_id, listing_id, scout_id
- inspection_metadata (timestamp_utc, geofence_verified, gps_coordinates, what3words, device_telemetry)
- power_infrastructure (system_type, generator_details, solar_inverter_details, trip_test_result, awarded_badge)
- internet_connectivity (isp_name, connection_type, plan_type, router_ups_present, speed_tests, awarded_badge)
- water_and_utilities (water_backup_present, source, storage_capacity_litres, filtration_installed)
- security_and_access (manned_gate_247, electric_perimeter_fence, smart_lock_functional, indoor_camera_detected, awarded_badge)
- cryptographic_verification (scout_public_key, payload_sha256_hash, scout_digital_signature)

Verification Flow

1. Scout completes audit checklist in app
2. App generates JSON payload from all captured data
3. App computes SHA-256 hash of payload
4. App signs hash with Ed25519 private key (stored in secure enclave)
5. Payload + signature uploaded to server
6. Server verifies:
 - Signature valid against scout's registered public key
 - Geofence: GPS coordinates within 50m of listing
 - Timestamp: Within 4 hours of audit start
 - Photo integrity: EXIF hashes match uploaded files
7. If all checks pass, payload forwarded to Badge Engine
8. Badge Engine evaluates against badge criteria and awards/denies

10.2 Scout Compensation Model

Activity	Payment	Timeline
Audit completion	C25-50 per audit (based on distance + complexity)	24 hours post verification
Re-audit (annual)	C15-30	Same
Dispute mediation	C10 per case	Post-resolution
Referral bonus (landlord signs up)	C20	Listing goes live
Quality bonus (zero disputes for 30 days)	C100	Monthly

11. ESCROW, PAYMENTS & MULTI-RAIL FINANCIAL ARCHITECTURE

11.1 Escrow Policies

Short Stays (< 30 days)

- 100% of booking capital held in escrow until 24 hours post check-in
- If no critical dispute raised, host payout dispatched instantly via MoMo/bank rails
- Guest cancellation: Full refund if >48hrs before check-in; 50% refund if 24-48hrs; no refund if <24hrs

Medium/Long Stays (1-12 months)

- First month's rent + security deposit held in escrow
- Ongoing rent disbursed on recurring monthly cycle via:
 - Automated MoMo debit (tenant authorizes once)
 - Employer payroll deduction
 - Flex-Advance circle payout
- Security deposit released within 14 days of move-out, minus damages (with photo evidence)

11.2 Multi-Rail Settlement Pattern

PAYMENT INGESTION Tenant pays via: MTN MoMo / M-Pesa / Card / Bank / USDC | v FLEX-LIVING ESCROW

- Funds held in segregated accounts per country
- Daily reconciliation across all rails
- Real-time balance tracking per booking | v PAYOUT DISBURSEMENT
- Host receives via preferred rail (MoMo default)
- Diaspora landlords receive USDC (auto-conversion)
- Flex Guides paid via MoMo
- Platform fees deducted automatically

11.3 USDC Stablecoin Rail (Diaspora)

Step	Action	Actor	Time
1	Diaspora investor deposits USDC to Flex-Living wallet	Investor	<10 min
2	Platform holds USDC in escrow	Flex-Living	Instant
3	Tenant pays monthly rent in local currency (GHS/NGN/KES)	Tenant	Monthly
4	Platform converts local currency to USDC at market rate	Flex-Living	Auto
5	USDC credited to investor wallet	Platform	Monthly
6	Investor withdraws USDC to personal wallet	Investor	<24 hrs

FX Risk Mitigation:

- Platform hedges 30-day forward contracts for major currency pairs
- Spread markup of 1.5% on conversion (vs. 5% bank spread)
- USDC held in Circle-backed custody (regulated, audited)

11.4 Employer Payroll Deduction

Integration Flow:

1. Employer signs partnership agreement with Flex-Living
2. HR platform integration (API or file-based)
3. Employee enrolls in Flex-Advance during onboarding
4. Monthly rent deducted pre-tax or post-tax from salary
5. Deduction report sent to Flex-Living
6. Platform credits tenant account and disburses to host

Benefits to Employer:

- Employee retention tool (housing stability reduces turnover)
- No cost to employer (Flex-Living pays integration fee)
- Branded as “employee wellness benefit”

12. SLA & AUTOMATED DISPUTE ENGINE

12.1 Outage Reporting Flow

[Tenant taps “Report Outage” in app] | [App presents categories: Power, Wi-Fi, Water, Security, Lock Access] | [App runs automated diagnostic based on category] |– Power: IoT smart plug ping + ambient light video capture + router offline check |– Wi-Fi: In-app multi-thread speed test to nearest cloud edge |– Water: Manual photo upload (no IoT yet) |– Security/Lock: Manual photo/video upload | [If diagnostic confirms breach against badge SLA] | [System auto-classifies severity tier] |– Tier 1 (Critical): Total infrastructure collapse |– Tier 2 (Degraded): Partial failure against badge promise |– Tier 3 (Minor): Delayed or reduced service | [Cure timer starts: 2hrs (T1), 4hrs (T2), 6hrs (T3)] | [Auto-alerts sent to host: SMS + WhatsApp + IVR voice call] | [Host receives countdown timer link] | [Host can: Upload resolution proof, request extension (once), or do nothing] | [Timer expires] | [Escrow auto-triggers refund per tier matrix] | [If Tier 1: 1-Tap Relocation Protocol activated] | [Guest receives: Refund credit + relocation options + free move]

12.2 Financial Compensation Matrix

Severity Tier	Triggering Event	Host Cure Window	Guest Remedy & Escrow Action	Host Penalty
Tier 1: Critical Infrastructure Collapse	Total power blackout with zero backup >1 hr; no water >3 hrs; total internet blackout >4 hrs	2 Hours	Immediate relocation to equivalent badged property + 100% refund of remaining unused nights + free ride credit	Badge stripped; \$50 relocation fine deducted from host balance; 1 strike added
Tier 2: Degraded Capacity	AC offline during power cut (on 24/7 Unbroken badge); Wi-Fi speeds <50% of badge tier for >3 hrs	4 Hours	30% daily rate refund credited from escrow for each affected 24-hour cycle	Badge downgraded to next lower tier; warning strike logged
Tier 3: Minor Disruption	Manual generator start delay (15-45 min); smart lock battery depleted; hot water heater failure	6 Hours	15% daily rate refund credited from escrow for that day	Mandatory maintenance receipt upload within 48 hours

12.3 Host Strike System

Strikes	Consequence
0	Full platform privileges
1	Warning + mandatory re-audit within 30 days
2	Listing suppressed in search (bottom 20% of results)

Table 46 – continued

Strikes	Consequence
3	Account suspension + escrow hold until all disputes resolved
4+	Permanent ban + legal referral for fraud

Strikes decay: 1 strike removed every 12 months of clean record.

13. MOVE-IN & LOGISTICS CONCIERGE

13.1 Service Overview

The Move-In Concierge handles the physical relocation of tenants from their old residence to their new Flex-Living verified property. This includes vehicle dispatch, condition inventory, and utility token hand-over.

13.2 Vehicle Fleet Model

Rather than owning vehicles, Flex-Living operates a **vett**ed driver marketplace:

Vehicle Type	Capacity	Use Case	Price Model
Bike/Courier	2 bags	Single room, light luggage	Fixed: C30-50 per trip
Small Van	1 room worth	Studio, single person	Fixed: C80-150 based on distance
Medium Truck	2-3 rooms worth	Apartment, family	Fixed: C200-350 based on distance + load
Large Truck	Full house	House, multi-bedroom	Fixed: C400-700 based on distance + load

13.3 Driver Vetting Process

1. **Application:** Driver submits ID, vehicle registration, insurance
2. **Background Check:** Criminal record check via local police
3. **Vehicle Inspection:** Flex Guide inspects vehicle condition, capacity, safety
4. **Test Run:** Driver completes 3 supervised moves with experienced Flex Guide
5. **Onboarding:** Driver receives branded vest, app access, GPS tracker
6. **Ongoing:** Monthly re-inspection, customer rating threshold (>4.0/5.0)

13.4 Move Job Lifecycle

[Booking Confirmed] | [Tenant requests Flex-Move during booking or later] | [App calculates fixed quote based on:] |– Distance (pickup to delivery, via What3Words) |– Load estimate (number of rooms, tenant self-reports) |– Vehicle type (recommended based on load) |– Time of day (peak hours surcharge: +20%) | [Tenant confirms quote and schedules date/time] | [Payment held in escrow (part of booking escrow

or separate)) | [System dispatches nearest available vetted driver] | [Driver arrives at pickup location] | [Condition Inventory: Driver + Tenant photograph all items before loading] | [Items loaded, GPS tracking activated] | [Driver delivers to new Flex-Living property] | [Condition Inventory: Driver + Tenant photograph all items after unloading] | [Tenant confirms delivery in app] | [Escrow released to driver (minus 15% platform commission)] | [Utility Token Handover:] |– ECG/Ikeja Electric/KPLC meter token auto-transferred |– Wi-Fi credentials shared via app |– Water tank level documented

13.5 Women's Safety Protocol

- Female tenants can **request female drivers** (if available in area)
- All drivers have **real-time GPS tracking** shared with tenant + emergency contact
- **SOS button** in app triggers immediate alert to Flex-Living ops + local police
- Driver identity (photo, ID number, vehicle plate) shared before pickup
- **Community verification:** Driver must be known/respected in their operating area

14. SECURITY, COMPLIANCE & REGULATORY FRAMEWORK

14.1 Data Protection & Privacy

Requirement	Implementation
Ghana Data Protection Act 2012	Consent management, data minimization, right to deletion
Nigeria NDPR (2019)	Cross-border transfer restrictions, DPO appointment
Kenya Data Protection Act 2019	Consent for biometric data, data subject rights
GDPR (for diaspora users)	Lawful basis, data portability, right to erasure

Technical Measures:

- Encryption at rest: AES-256 for all databases
- Encryption in transit: TLS 1.3 for all APIs
- PII tokenization: Phone numbers and IDs hashed in logs
- Data retention: Audit logs 7 years, user data 3 years post-deletion request
- Access control: RBAC with principle of least privilege

14.2 Financial Compliance

Jurisdiction	Requirement	Flex-Living Approach
Ghana	Bank of Ghana payment services regulation	Partner with licensed payment provider; don't hold funds directly
Nigeria	CBN mobile money guidelines	Flutterwave/Paystack handle NGN custody

Table 49 – continued

Jurisdiction	Requirement	Flex-Living Approach
Kenya	CBK National Payment System Act	M-Pesa Daraja integration via licensed partner
All	Anti-Money Laundering (AML)	KYC Tier 2+ for transactions >\$1,000; suspicious activity reporting
All	FATF Travel Rule (for USDC)	Transaction monitoring for crypto rails >\$1,000

14.3 Property & Rental Law Compliance

Market	Legal Framework	Compliance Action
Ghana	Rent Act 1963 (Act 220); REAC Act 2020	Full REAC licensing for agents; advance capped at 6 months (platform enforces)
Nigeria	State-level tenancy laws (Lagos Tenancy Law 2011)	Digital lease templates compliant per state; Lagos priority
Kenya	Landlord and Tenant (Shops, Hotels and Catering Establishments) Act	Standard lease terms; deposit protection
All	Contract law	Digital signatures legally binding; e-signature act compliance

14.4 Tax Compliance Automation

Jurisdiction	Tax Type	Automation
Ghana	Rental income tax (8% flat for residential)	Auto-calculation from escrow payouts; quarterly GRA filing
Nigeria	Personal income tax (state-level)	Auto-calculation; state-appropriate filing
Kenya	Withholding tax on rent (10% for residents, 30% non-resident)	Auto-deduction from diaspora payouts; KRA filing
All	VAT (where applicable)	Auto-calculation on platform fees; monthly filing

Landlord Dashboard Feature:

- Quarterly tax estimate visible in real-time
- One-tap filing (pre-filled forms)
- Payment confirmation and receipt storage
- Annual tax summary for personal filing

15. INFRASTRUCTURE & DEVOPS STACK

15.1 Cloud Architecture

Layer	Technology	Justification
Primary Cloud	AWS (us-east-1 + af-south-1)	Global reach + African latency optimization
Disaster Recovery	Multi-region active-passive	RPO < 5 min, RTO < 30 min
CDN	CloudFront + Cloudflare	Static assets, media delivery, DDoS protection
Container Orchestration	Amazon EKS (Kubernetes)	Microservices scaling, self-healing
Serverless	AWS Lambda (event processing)	Cost-effective for sporadic workloads

15.2 Database & Storage

Service	Technology	Use Case
Primary Database	PostgreSQL 15 + PostGIS	Relational data, spatial queries
Read Replicas	PostgreSQL (multi-AZ)	Search, reporting, analytics
Cache	Redis Cluster (ElastiCache)	Sessions, rate limiting, real-time data
Search	Elasticsearch	Listing search, geo-spatial queries, badge filtering
Object Storage	S3 + CloudFront	Photos, videos, audit evidence
Backup	S3 Glacier + cross-region replication	7-year retention for audit logs
Time-Series	InfluxDB (self-hosted)	IoT telemetry, speed test history

15.3 Message Bus & Event Streaming

Service	Technology	Use Case
Primary Message Bus	Apache Kafka (MSK)	Cross-service async communication
Dead Letter Queue	AWS SQS	Failed event retry
Event Store	Kafka (compacted topics)	Audit trail, event sourcing
Real-Time	WebSocket (API Gateway)	Live tracking, notifications

15.4 CI/CD & DevOps

Practice	Tool	Configuration
Source Control	GitHub	Monorepo with service boundaries
CI/CD	GitHub Actions	Build, test, deploy pipelines
Infrastructure as Code	Terraform	All AWS resources codified
Container Registry	Amazon ECR	Immutable image tags
Monitoring	Datadog + AWS CloudWatch	APM, logs, metrics, alerts

Table 55 – continued

Practice	Tool	Configuration
Error Tracking	Sentry	Real-time error alerting
API Documentation	Swagger/OpenAPI 3.0	Auto-generated from code

15.5 Security Stack

Layer	Tool/Practice
WAF	AWS WAF + Cloudflare
DDoS Protection	Cloudflare Magic Transit
Secrets Management	AWS Secrets Manager + HashiCorp Vault
Vulnerability Scanning	Snyk (dependencies) + Trivy (containers)
Penetration Testing	Quarterly by third-party (Cobalt.io)
Compliance Monitoring	Vanta (SOC 2 prep)

15.6 Mobile Development

Platform	Framework	Offline Strategy
Tenant App	React Native	Redux Persist + SQLite; sync queue
Landlord App	React Native	Same
Flex Guide App	React Native	SQLite + background sync; works fully offline
Minimum OS	Android 8.0 / iOS 13	Covers 95% of African smartphones

15.7 Web Development

Platform	Framework	Hosting
Tenant Web	Next.js 14	Vercel (edge)
Landlord Dashboard	Next.js 14	Vercel
Diaspora Dashboard	Next.js 14	Vercel
Admin Dashboard	Next.js 14	Vercel
WhatsApp Bot	Node.js + Express	AWS Lambda
USSD Gateway	Node.js + Express	AWS Lambda

16. IMPLEMENTATION ROADMAP

16.1 Phase 1: Foundations (Weeks 1-4)

Week	Deliverable	Owner	Success Criteria
1	Multi-tenant DB schema (PostgreSQL + PostGIS); Auth pipeline (phone OTP)	Backend	DB migrations clean; OTP delivery <30s
2	KYC integration (Ghana Card API, Smile Identity); User roles	Backend	KYC Tier 1 flow complete; selfie match >95% accuracy
3	Listing Catalog Service; What3Words + Plus Codes integration; Elasticsearch indexing	Backend	Search returns results <500ms; geo-queries accurate
4	Basic tenant mobile app (browse, search, view listing); Landlord WhatsApp bot v1	Mobile + Bot	App installable; bot responds to “Hi” with onboarding flow

16.2 Phase 2: Field Scout Engine (Weeks 5-8)

Week	Deliverable	Owner	Success Criteria
5	React Native Flex Guide app scaffold; offline SQLite cache; camera integration	Mobile	App installs on Android 8+; photos save locally
6	GPS geofencing; EXIF hashing; custom speed test SDK	Mobile	Geofence accuracy <10m; speed test matches Ookla
7	Ed25519 key generation in secure enclave; payload signing; upload/sync	Mobile + Backend	Signature verifies server-side; offline sync <5s
8	Scout Dispatch Dashboard; QA review workflow; badge assignment engine	Backend + Admin	Dispatch algorithm assigns nearest scout; badge rules engine accurate

16.3 Phase 3: Payments & Escrow (Weeks 9-12)

Week	Deliverable	Owner	Success Criteria
9	MTN MoMo + Telecel Cash webhooks; Paystack integration	Backend	Payment confirmation <10s; webhook idempotency tested
10	Escrow State Machine; hold/release/freeze states; automated payouts	Backend	State transitions atomic; payout settles <24hrs
11	M-Pesa Daraja integration; Flutterwave multi-currency; USDC deposit address generation	Backend	All rails tested in sandbox; USDC deposit confirms on-chain
12	Financial reconciliation dashboard; daily settlement reports	Admin	Reports accurate to the cent; exception handling tested

16.4 Phase 4: Guest Experience & SLA (Weeks 13-16)

Week	Deliverable	Owner	Success Criteria
13	Tenant booking flow (search -> book -> pay); Flex-Stay pricing (daily/weekly/monthly)	Mobile	End-to-end booking <3 min; pricing calculations correct
14	Digital lease generation + e-signature; host approval workflow	Backend	Lease legally compliant; e-signature binding
15	In-app outage reporting; automated diagnostics (speed test, IoT ping); cure timer	Mobile + Backend	Report -> diagnostic -> alert flow <60s
16	Auto-refund engine; 1-Tap Relocation Protocol; host strike system	Backend	Refund triggers correctly; relocation options relevant

16.5 Phase 5: Logistics & Pilot Launch (Weeks 17-20)

Week	Deliverable	Owner	Success Criteria
17	Flex-Move driver marketplace; fixed pricing algorithm; GPS tracking	Backend + Mobile	Quote accurate within 10%; tracking updates every 30s
18	Condition inventory photo logging; check-in/check-out workflow	Mobile	Before/after photos timestamped and hashed
19	Utility token API integration (ECG, Ikeja Electric, KPLC); auto-handover	Backend	Token transfer successful; credentials shared securely
20	CLOSED BETA LAUNCH —Accra + Lagos; 20 Flex Guides; 100 direct hosts; corporate/diaspora users	All	500 verified listings; 200 successful bookings; <5% dispute rate

16.6 Post-MVP Roadmap

Quarter	Focus	Key Deliverables
Q3 2026	Ghana Scale	5,000 listings; Flex-Advance public launch; 50 Flex Guides
Q4 2026	Nigeria Entry	Lagos launch; 15,000 listings; Paystack primary rail
Q1 2027	Kenya Entry	Nairobi launch; M-Pesa integration; 10,000 listings
Q2 2027	Diaspora Suite	USDC full flow; London/NYC marketing; 500 diaspora landlords
Q3 2027	API Ecosystem	Flex Score API; 20 fintech partners; B2B revenue stream

Table 64 – continued

Quarter	Focus	Key Deliverables
Q4 2027	Pan-African	8 countries; 200,000 users; profitability

17. RISK REGISTER & MITIGATION

Risk ID	Risk	Probability	Impact	Mitigation	Owner
R-01	Landlords refuse to list without agent intermediaries	Medium	High	Scout-mediated onboarding; “we bring tenants, you keep full rent” pitch	Growth
R-02	Flex Guide bribery by landlords to fake audits	Medium	Critical	Cryptographic signing (can’ t fake without private key); neighbor verification; random re-audits	Product
R-03	RentSusu/Flex-Advance mass defaults	Medium	Critical	Employer payroll deduction tier; default insurance (5% of pool); credit bureau integration; conservative initial circles	Finance
R-04	Mobile money API downtime	High	High	Multi-rail fallback (if MoMo fails, auto-switch to Paystack/card); offline payment queue	Engineering
R-05	USDC regulatory ban	Low	High	MoMo/card rails as primary; USDC positioned as “diaspora convenience” not core; legal monitoring	Legal
R-06	Competitor price war (PropertyPro.ng free listings)	Medium	Medium	Zero commission is already our model; differentiation is trust/financing, not price	Strategy

Table 65 – continued

Risk ID	Risk	Probability	Impact	Mitigation	Owner
R-07	IoT smart plug theft/damage	Medium	Medium	Landlord-owned (not platform-owned); insurance rider; deposit deduction for damage	Operations
R-08	Data breach exposing KYC documents	Low	Critical	Encryption at rest + transit; minimal PII storage; quarterly pen tests; SOC 2 certification	Security
R-09	Scout safety during property visits	Medium	High	GPS tracking shared with ops; SOS button; visit scheduling during daylight; high-crime area flagging	Operations
R-10	Government regulatory shutdown	Low	Critical	Full compliance from Day 1; government partnerships; legal counsel in each market; trade association membership	Legal
R-11	Currency devaluation eroding escrow balances	High	Medium	Daily reconciliation; USD-peg option for diaspora; hedging for large balances	Finance
R-12	Tenant fraud (fake identity, payment reversals)	Medium	High	KYC Tier 2+ for long stays; mobile money irreversibility (where supported); Flex Score threshold for premium properties	Trust

18. SUCCESS METRICS & KPIs

18.1 North Star Metric

Monthly Verified Bookings (MVB): Number of bookings completed where both listing verification and tenant satisfaction thresholds are met.

Target: 10,000 MVB by Month 18.

18.2 Primary KPIs

KPI	Definition	Target (M6)	Target (M12)	Target (M18)
Verified Listings	Live listings with >=1 active badge	500	5,000	50,000
Active Users	MAU (monthly active users)	1,000	10,000	50,000
Booking Conversion	Search -> Booking rate	8%	12%	15%
Listing Accuracy	% of listings matching scout report (dispute rate inverse)	99.5%	99.7%	99.8%
Flex-Advance Uptake	% of long-stay bookings using Flex-Advance	20%	40%	50%
Scout Network	Active Flex Guides	20	50	200
GMV	Gross Merchandise Value (total booking value)	\$50K/mo	\$500K/mo	\$5M/mo
Net Revenue	Platform revenue after payouts	\$5K/mo	\$50K/mo	\$500K/mo
NPS	Net Promoter Score (tenants)	+30	+40	+50
Organic Acquisition	% of new users from referrals/organic	60%	65%	70%
Diaspora Landlords	Active diaspora-managed properties	0	50	500

18.3 Operational Metrics

Metric	Target	Measurement
Scout audit completion time	<48 hours from request	Ticket timestamp -> report submission
Escrow payout speed	<24 hours post check-in	Check-in confirmation -> host wallet credit
Outage resolution rate	>80% within cure window	Outage claims resolved by host before timer expiry
Move job on-time rate	>95% within scheduled window	Scheduled time -> actual arrival

Table 67 – continued

Metric	Target	Measurement
Customer support response	<2 hours for chat, <24 hours for email	First response time
App crash rate	<0.5% of sessions	Sentry error tracking
API uptime	>99.9%	CloudWatch monitoring

19. APPENDICES

Appendix A: Glossary

Term	Definition
Flex Guide	Community scout who verifies property infrastructure and assigns badges
Flex-Advance	Rotating savings circle or employer payroll deduction for rent advance financing
Flex Score	Rental credit score (300-850) based on payment history, reviews, and verification
Flex-Move	On-demand moving and relocation service booked through platform
Badge	Infrastructure certification awarded after scout audit (power, internet, security, etc.)
Escrow	Third-party holding of funds until contractual conditions are met
MoMo	Mobile Money (MTN Mobile Money, M-Pesa, etc.)
USDC	USD Coin stablecoin on Solana/Polygon blockchains
What3Words	Geolocation system dividing world into 3m x 3m squares with unique 3-word addresses
REAC	Real Estate Agency Council (Ghana)

Appendix B: Technology Stack Summary

Layer	Technology
Mobile Apps	React Native
Web Frontend	Next.js 14
API Backend	Node.js (Express/NestJS), Python (FastAPI), Go
Databases	PostgreSQL 15 + PostGIS, Redis, Elasticsearch, InfluxDB
Message Bus	Apache Kafka (AWS MSK)
Cloud	AWS (primary), Cloudflare (CDN/WAF)
Containers	Docker, Kubernetes (EKS)
CI/CD	GitHub Actions, Terraform

Table 69 – continued

Layer	Technology
Monitoring	Datadog, CloudWatch, Sentry
Crypto	Ed25519 (signing), SHA-256 (hashing), Solana/ Polygon (USDC)

Appendix C: External API Integrations

Service	API	Use Case	Status
MTN MoMo	REST API	Mobile money payments	Required
M-Pesa Daraja	REST API	Mobile money payments (Kenya)	Required
Paystack	REST API	Card/bank payments (Nigeria)	Required
Flutterwave	REST API	Multi-currency card/bank	Required
What3Words	REST API	Addressing and navigation	Required
Google Maps	REST/JS API	Routing, geocoding	Required
Smile Identity	REST API	KYC/ID verification	Required
Circle (USDC)	REST API	Stablecoin deposits/withdrawals	Phase 3
Twilio	REST API	SMS, IVR, WhatsApp Business	Required
Africa' s Talking	REST API	SMS (alternative to Twilio)	Backup
ECG/Ikeja Electric/KPLC	REST API	Utility token purchase/transfer	Phase 5

Appendix D: Document Control

Version	Date	Author	Changes
1.0.0-MVP	Original	PRD Author	Initial technical specification
2.0.0	August 2026	Flex-Living Strategy Team	Fused market strategy, business model, community growth, and technical PRD into unified build document

END OF DOCUMENT

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